

COMICCRAFT – PROJECT DOCUMENTATION

AI COMIC STORY CREATOR USING GEMINI MODELS

1. Project Overview

ComicCraft is a web-based Generative AI application designed to create personalized comic stories from user-provided ideas.

The application combines Artificial Intelligence, web technologies, image generation, comic layout processing, and PDF generation to provide an end-to-end comic creation workflow.

2. Problem Statement

Creating a complete comic manually requires considerable time and effort.

Users need to:

- Develop a story
- Create characters
- Write dialogues
- Design scenes
- Create images
- Arrange comic panels
- Prepare the final document

ComicCraft addresses this problem by automating major parts of the comic creation process using Generative AI.

3. Project Objective

The main objectives of ComicCraft are:

- Generate comic stories from simple user ideas.
- Use Gemini AI for story generation.
- Generate structured comic panels.
- Create visual content for panels.
- Combine text and images into a comic layout.
- Export the final comic as a PDF.
- Provide a simple and user-friendly interface.

4. Proposed Solution

ComicCraft provides a web-based platform where users can enter a story idea.

The application processes the prompt using AI and generates:

- Story outline
- Detailed comic panels
- Scene descriptions
- Captions
- Narration
- Dialogue

- Image prompts
- Panel images

The generated content is then assembled into a comic and exported as a PDF.

5. Technology Stack

The project uses the following technologies:

Frontend

- HTML
- CSS
- Jinja2 templates

Backend

- Python
- FastAPI

Artificial Intelligence

- Gemini Generative AI models

Image Processing

- Image generation module
- Pillow

PDF Generation

- FPDF / compatible PDF generation library

Development Tools

- Visual Studio Code
- Python virtual environment
- Web browser
- GitHub for project version control

6. System Architecture

The major components of ComicCraft are:

1. User Interface
2. FastAPI Backend
3. Gemini AI
4. Image Generation Module
5. Comic Layout Module
6. PDF Generation Module

Architecture Flow

User

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Web Interface

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FastAPI Backend

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Gemini AI

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Story Panels + Image Prompts

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Image Generation

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Comic Layout

↓

PDF Generation

↓

Final Comic

7. Project Workflow

The complete project workflow is:

Step 1 – User Input

The user enters a story idea through the web interface.

Step 2 – Prompt Validation

The backend checks whether the prompt is valid and suitable for processing.

Step 3 – Story Generation

Gemini AI generates the story outline and detailed comic panels.

Step 4 – Image Prompt Generation

Each comic panel contains an image prompt describing the required visual scene.

Step 5 – Image Generation

The image generation module processes the image prompts and creates panel images.

Step 6 – Comic Layout

The story content and generated images are combined into a structured comic layout.

Step 7 – PDF Generation

The completed comic layout is converted into a PDF document.

Step 8 – Final Output

The user can view or access the generated comic and PDF.

8. Major Modules

8.1 Frontend Module

Provides the user interface for entering prompts and displaying results.

8.2 Backend Module

Handles requests, validation, AI processing, image processing, layout creation, and PDF generation.

8.3 Gemini AI Module

Generates story outlines and detailed comic panel information.

8.4 Image Generation Module

Creates or processes visual images for comic panels.

8.5 Comic Layout Module

Combines text and images into an organized comic structure.

8.6 PDF Generation Module

Converts the completed comic layout into a PDF file.

9. Functional Requirements

The system should:

- Accept a story prompt.
- Validate user input.
- Generate an AI-based story.
- Create multiple comic panels.
- Generate image prompts.
- Generate or process panel images.
- Build the comic layout.
- Generate a PDF.
- Provide suitable error messages.

10. Non-Functional Requirements

The system should provide:

- Usability
- Reliability

- Maintainability
- Security
- Compatibility
- Reasonable performance
- Modular architecture

Sensitive API credentials should be protected from public exposure.

11. Testing

The application is tested using:

- Functional testing
- Integration testing
- Input validation testing
- Error handling testing
- Application workflow testing

The testing process verifies the major components from user input to final PDF generation.

12. Deployment

ComicCraft can be run locally using a Python environment and FastAPI server.

The basic deployment process includes:

1. Set up the project environment.
2. Install required dependencies.
3. Configure API credentials.
4. Start the FastAPI application.

5. Open the application in a web browser.
6. Verify the application workflow.

13. Advantages

ComicCraft provides:

- Faster comic creation
- AI-assisted storytelling
- Automated panel generation
- Reduced manual effort
- Integrated text and image generation
- PDF export
- Simple user workflow

14. Future Enhancements

Future versions can include:

- Multiple comic art styles
- Better character consistency
- User-selected characters
- Multilingual story generation
- Voice-based story input
- User accounts
- Cloud deployment

- Online comic sharing
- Additional AI models
- Improved image generation

15. Conclusion

ComicCraft demonstrates how Generative AI can be integrated with web application technologies to create personalized digital comics.

The application combines AI story generation, image generation, comic layout, and PDF export into a single workflow.

The project provides a foundation for developing more advanced AI-assisted creative applications in the future.